

4 (a) force on proton is normal to velocity and field
provides centripetal force (for circular motion) M1
A1 [2]

(b) magnetic force = Bqv B1
centripetal force = $mr\omega^2$ or mv^2/r B1
 $v = r\omega$ B1
 $Bqv = Bqr\omega = mr\omega^2$
 $\omega = Bq/m$ A1 [4]

5 (a) either $\phi = BA \sin \theta$ M1